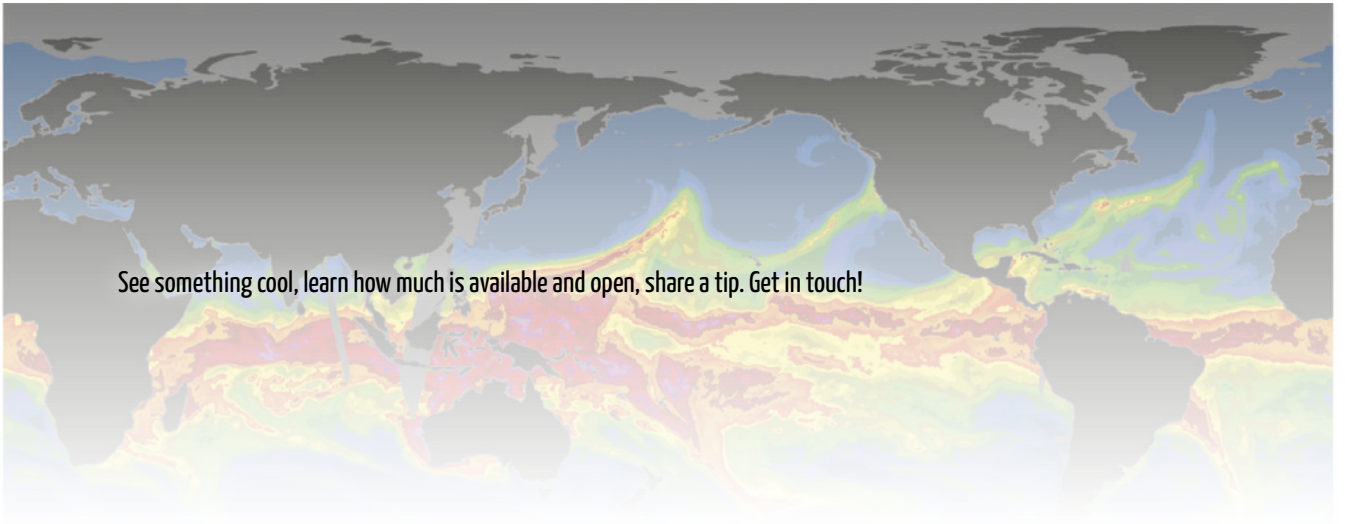
Empower Yourself With Data!



See something cool, learn how much is available and open, share a tip. Get in touch!

http://fuelscape.net/fellowship_data_carpentry.html#1

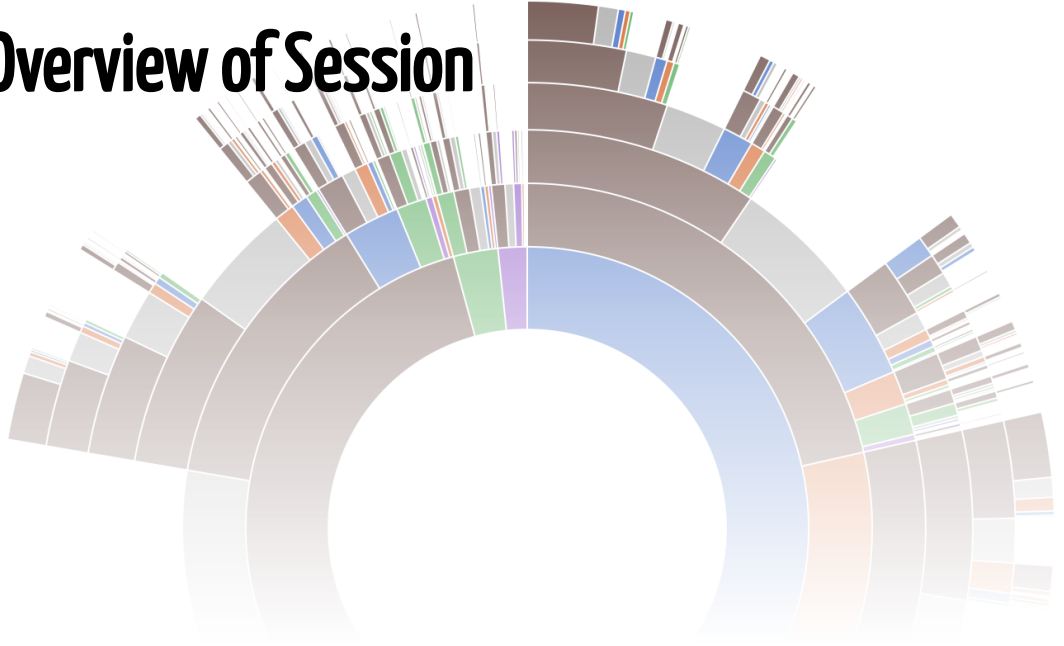
https://github.com/scotfree/fuelscape/

gateless@gmail.com

https://www.linkedin.com/in/scot-kennedy-7a04265/

Scot Kennedy

Overview of Session



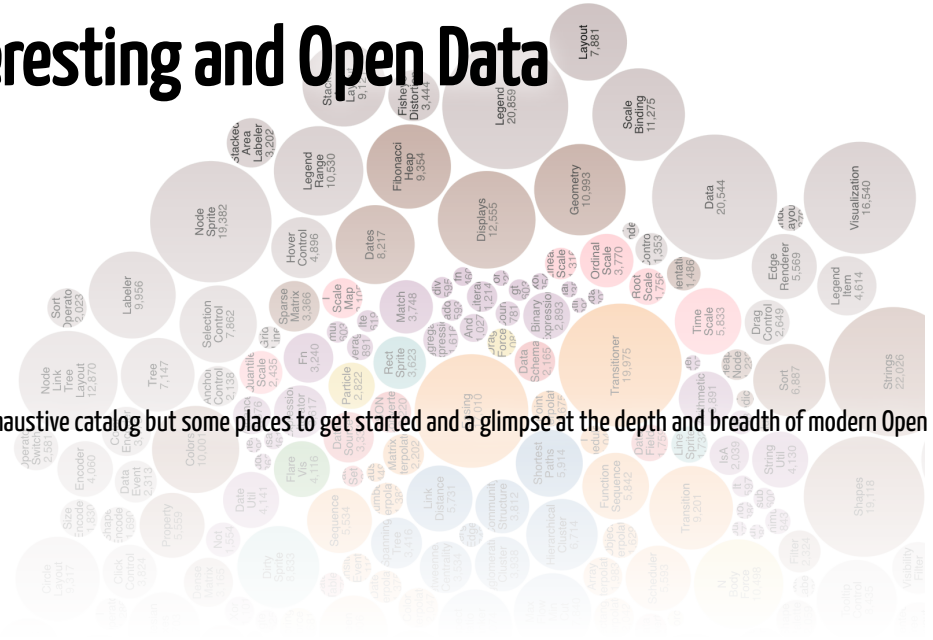
Interesting and **Open** Data

Useful and **Open** Tools

Opening PandorAI's Box

Open Horizon

Interesting and Open Data



Not an exhaustive catalog but some places to get started and a glimpse at the depth and breadth of modern Open Data.

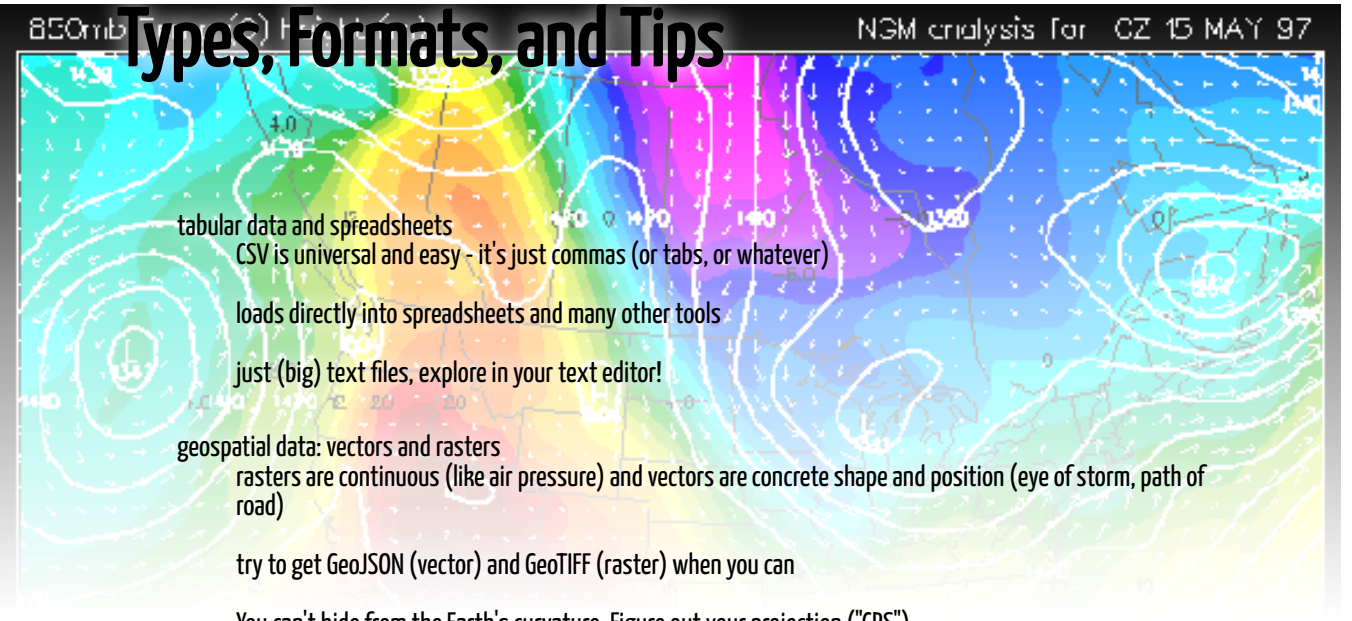
Types and Formats

Maps

Climate and Environmental Datasets

Cloud Data

Types, Formats, and Tips



tabular data and spreadsheets

CSV is universal and easy - it's just commas (or tabs, or whatever)

loads directly into spreadsheets and many other tools

just (big) text files, explore in your text editor!

geospatial data: vectors and rasters

rasters are continuous (like air pressure) and vectors are concrete shape and position (eye of storm, path of road)

try to get GeoJSON (vector) and GeoTIFF (raster) when you can

You can't hide from the Earth's curvature. Figure out your projection ("CRS")

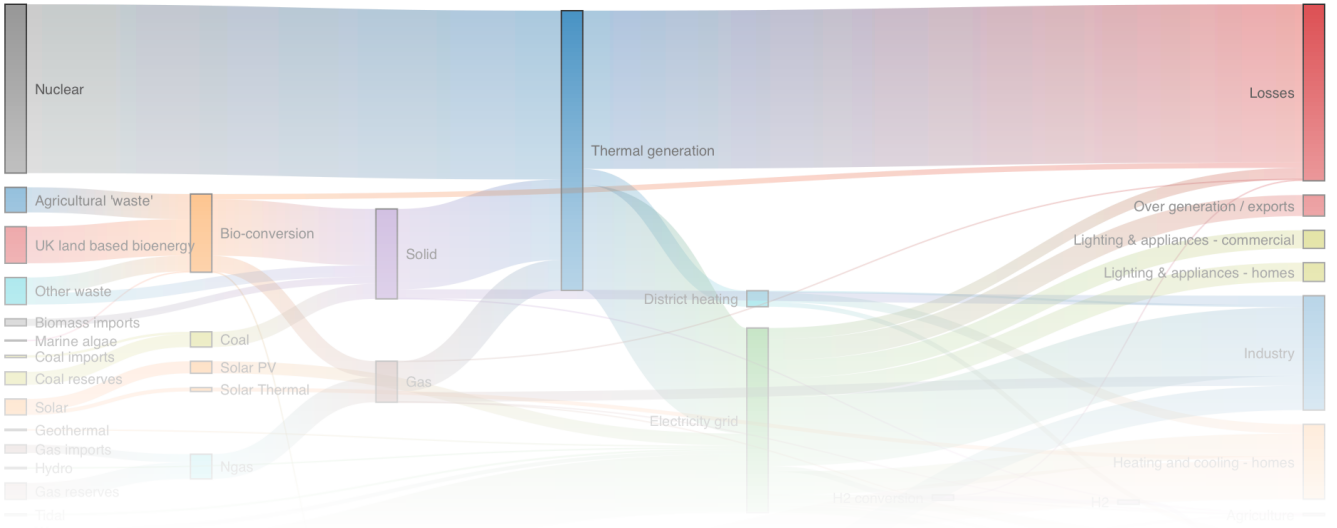
other

ZIP and TGZ are just archives, easy to unzip

JSON and YAML are wonderful and common, almost as easy to work with as CSV

"Dataframes" are like spreadsheets with lots of code attached and no built-in interface, widely usable in python and apps.

environmental and societal



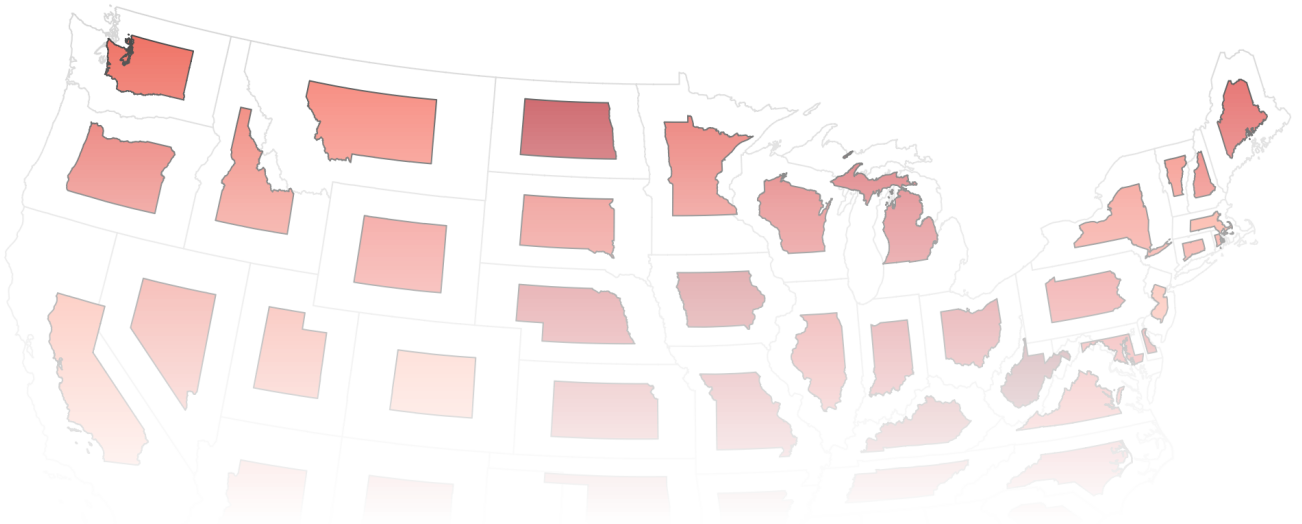
https://data.humdata.org

https://jeiloh.github.io/esip_dlc_trusted_data/

NASA hosts an incredible variety of climate-relevant datasets.

UN Food & Ag has an amazing range of global datasets.

Maps and Geospatial Data



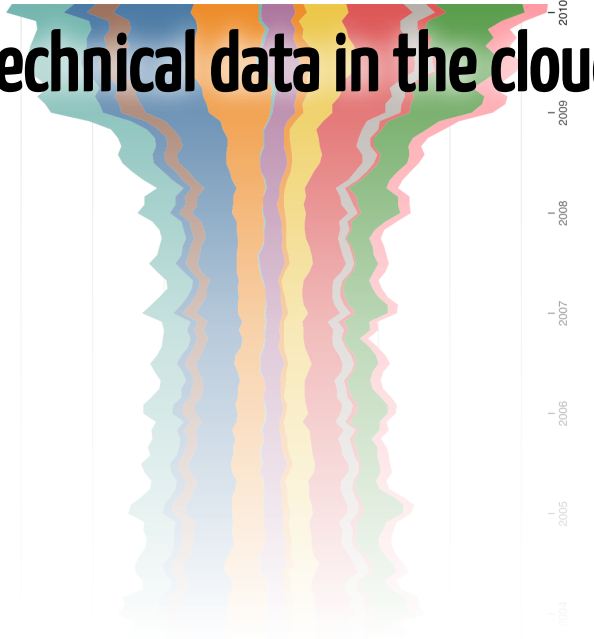
Open Street Map

- Simple Maps
- Turbo and Overpass

Overture

OpenTopography for landform and elevation data.

huge and technical data in the cloud



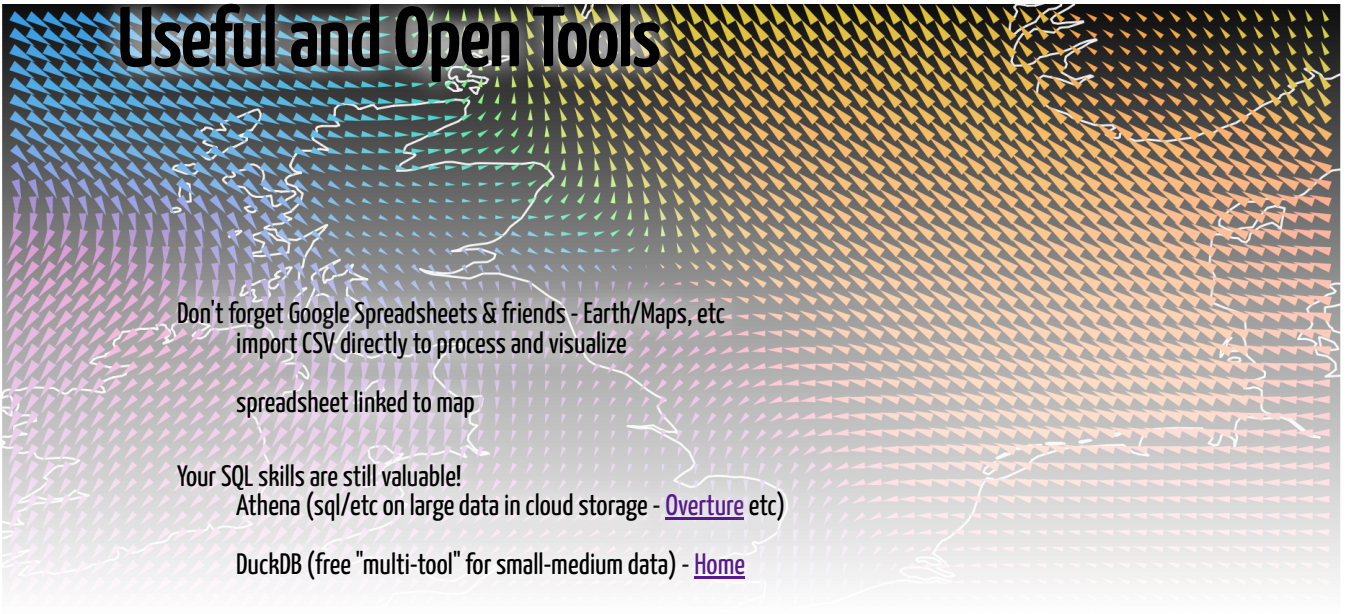
AWS Open Data

Planet Labs for non-profits

NOAA uses cloud platforms

Kaggle has a lot of random interesting datasets

Useful and Open Tools



Don't forget Google Spreadsheets & friends - Earth/Maps, etc
    import CSV directly to process and visualize

    spreadsheet linked to map

Your SQL skills are still valuable!
    Athena (sql/etc on large data in cloud storage - Overture etc)

    DuckDB (free "multi-tool" for small-medium data) - Home

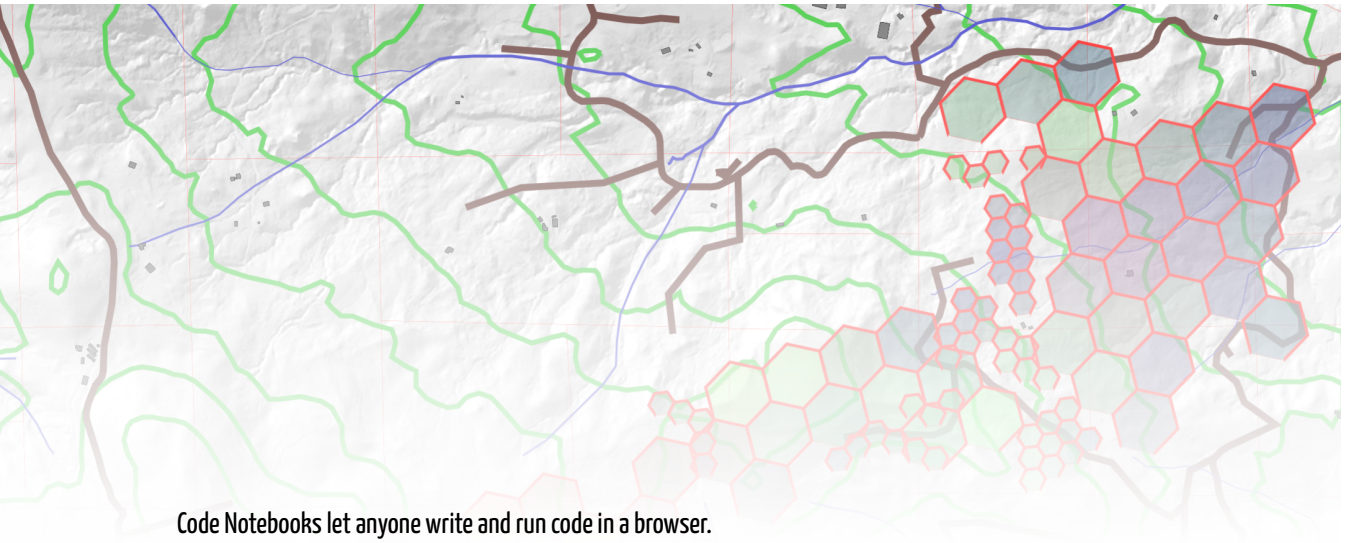
Geospatial Tools
    QGIS - Free and popular GIS

    geojson.io and Felt - look at and edit most geo data in a web page

    export data from OSM or USGS - check your local government! {Humboldt County Web GIS](https://cty-gis-web.co.humboldt.ca.us/HCEGIS3.0/)

Put some data in AWS S3 and be amazed! There's an analysis ecosystem already pointed at it!

Notebooks!



Code Notebooks let anyone write and run code in a browser.

Work from examples not write code.

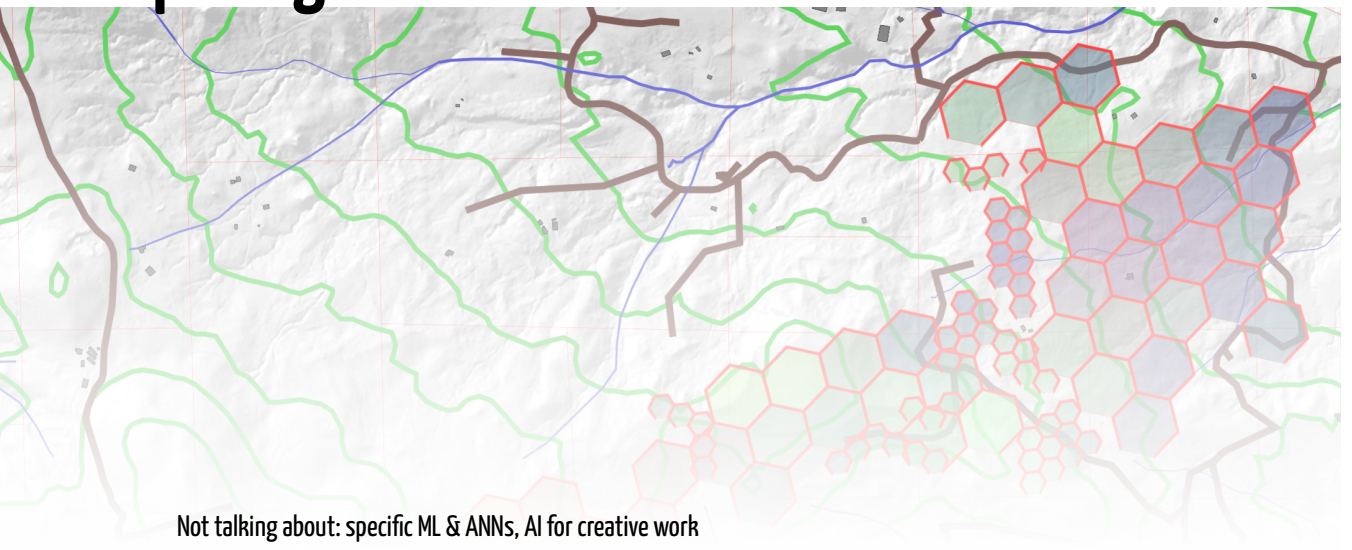
With a narrative direction, graphics, and typesetting they are built for storytelling. Matrix Math

Try them free on Google Colab (meh) or AWS (better).

Geospatial Notebook Collection is a great way to learn A LOT, especially about Geospatial data.

More examples

Opening PandorAI's box



Not talking about: specific ML & ANNs, AI for creative work

Try to make tools not answers

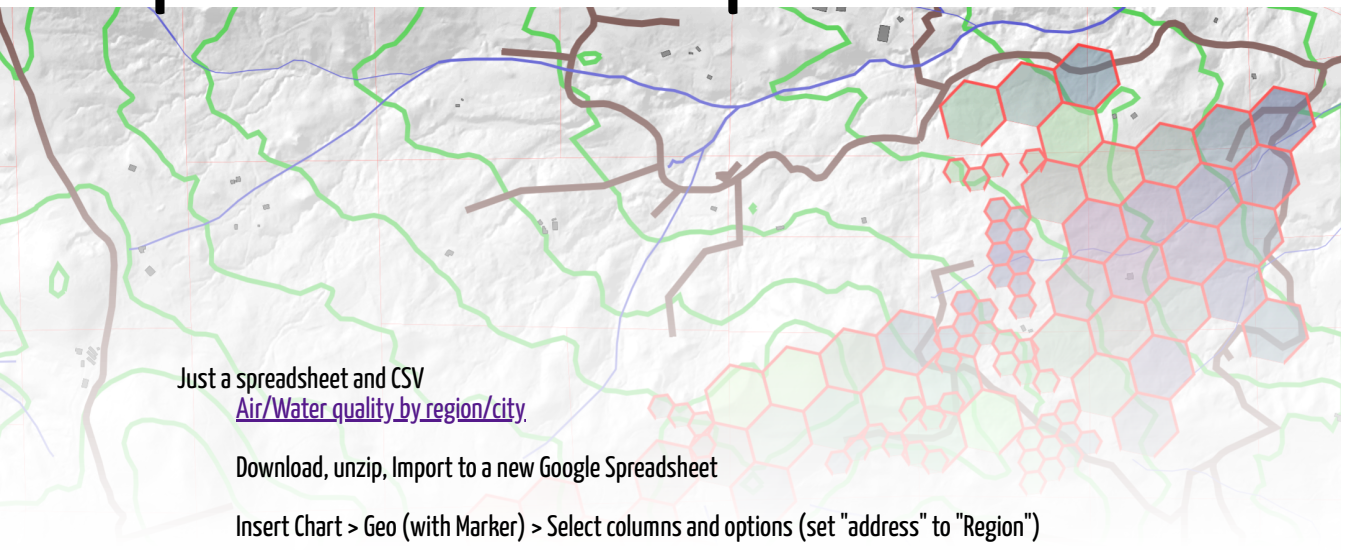
AI to make tools not analyze data - less waste and hallucination

Have it Ask You Questions

Cautionary Tale: Board Game Rules

Use AI to get you over the "energy barrier" for trying new things, not to do the work

Open Horizons and Examples



Just a spreadsheet and CSV
Air/Water quality by region/city

Download, unzip, Import to a new Google Spreadsheet

Insert Chart > Geo (with Marker) > Select columns and options (set "address" to "Region")

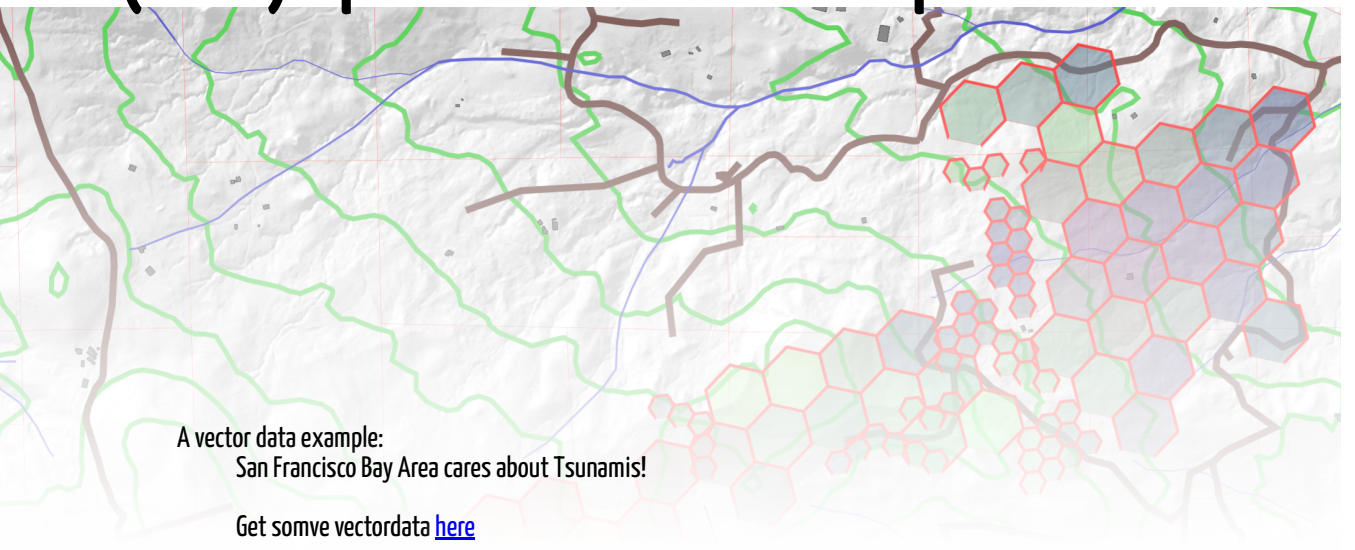
A raster/CSV data example:
Global "risk" data

Notebook with some basic Geospatial tools in place

Try adding another data set

or Energy Access

(more) Open Horizons and Examples



A vector data example:

San Francisco Bay Area cares about Tsunamis!

Get somve vectordata here

and look at it with geojson.io

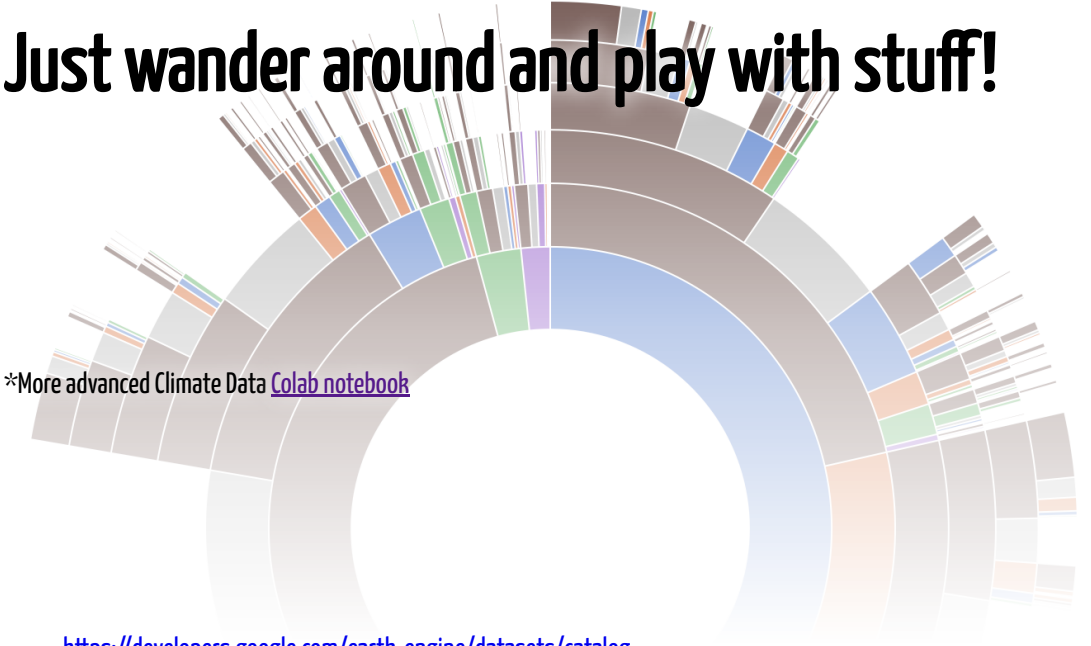
Do some visualization and modelbuilding!

simple data here

empty notebook

use AI to build your visualization!

Just wander around and play with stuff!



*More advanced Climate Data Colab notebook

https://developers.google.com/earth-engine/datasets/catalog

https://notebooks.gishub.org (start with "blog")

https://felt.com

https://www.epa.gov/enviroatlas/enviroatlas-dynamic-data-matrix

https://colab.research.google.com/github/neuromatch/climate-course-content/blob/main/tutorials/W1D3_RemoteSensing/student/W1D3_Tutorial4.ipynb